

Sealed Experiment Record

Sealed 2026-10-10 · Signed by the K-Veritas server
Session aa5c386e · Data hash 3ea1...9262

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-10 19:31 and 20:38 UTC. Reported results: final_eval_loss 3.8481, final_eval_perplexity 46.9039. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	aa5c386e-0266-41c8-9fa7-255bcbd67948
Initialized	2026-10-10T19:31:53Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	633d1e0bb131f814e067cde2d67e84a4a794d5122554628b17ad795e04008390
Disclosure	open
Source code, run 1	3f97ca364906ad444f57bf4c5f3a6bb226429b93dfd163aa01db577e6d458da9

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	1 hours 6 minutes	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
params	58.07		heuristic
enabled	25.3		heuristic
loss	3.8481		heuristic
eval_loss	3.8481	2000 (4 readings)	explicit

Metric	Final value	Step	Source
eval_perplexity	46.9039	2000 (4 readings)	explicit

Inline claim	Value	Phase	Output line
final_eval_loss	3.8481		21
final_eval_perplexity	46.9039		22

Seed committed: seed:0, output line 1, 2026-10-10T19:31:53Z.

4 Execution evidence

Coherence

Coherent (0.41): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	8619.1 s	16.48 GB	99%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 35 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	19:31:53	786520ed7292582e	35
2	run_end	20:38:33	786520ed7292582e	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 36 files read, 0 files written, 15 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-10T20:38:35.345788+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	3ea1024baf8c7e0c27e846061055683df0b20293833dcb6e85c6acd0ddb29262
Code bundle hash	8fff306ee6827640b6a7389baa870efaea748ec278598d61b62383cecd2114f6

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
params	58.07		4	heuristic
enabled	25.3		5	heuristic
loss	3.8481		12	heuristic
eval_loss	4.43855	500	13	explicit
eval_perplexity	84.6524	500	14	explicit
eval_loss	4.06779	1000	15	explicit
eval_perplexity	58.4275	1000	16	explicit
eval_loss	3.90539	1500	17	explicit
eval_perplexity	49.6695	1500	18	explicit
eval_loss	3.8481	2000	19	explicit
eval_perplexity	46.9039	2000	20	explicit

Phase	Line	Time	CPU time	Memory	GPU
pretrain	2	19:31:53	0.4 s	0.45 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	19:31:53	786520ed7292582e	35
1	phase pretrain	19:31:53	786520ed7292582e	0
2	run_end	20:38:33	786520ed7292582e	0

Event	Path or command	Hash
read	kv_run.sh	
read	configs/llama_100m.json	
read	configs/llama_130m.json	
read	configs/llama_1b.json	
read	configs/llama_20m.json	
read	configs/llama_250m.json	
read	configs/llama_350m.json	
read	configs/llama_35m.json	
read	configs/llama_3b.json	
read	configs/llama_40m.json	
read	configs/llama_60m.json	
read	configs/llama_71m.json	
read	configs/llama_7b.json	
read	configs/llama_9m.json	
read	galore_torch/adamw.py	
read	galore_torch/galore_projector_tensor.py	
read	peft_pretraining/dataloader.py	
read	run_glue.py	
read	scripts/benchmark_c4/llama_130m.sh	

Event	Path or command	Hash
read	scripts/benchmark_c4/llama_1b.sh	
read	scripts/benchmark_c4/llama_350m.sh	
read	scripts/benchmark_c4/llama_60m.sh	
read	scripts/benchmark_c4/llama_7b.sh	
read	scripts/single_gpu/llama_7b.sh	
read	scripts/single_gpu/llama_7b_checkpointing.sh	
read	scripts/tensor_test/neural_operator.py	
read	setup.py	
read	torchrun_main.py	
read	galore_torch.egg-info/PKG-INFO	
read	peft_pretraining/training_utils.py	
read	peft_pretraining/args_utils.py	
read	peft_pretraining/modeling_llama.py	
read	galore_torch/__init__.py	
read	galore_torch/adafactor.py	
read	galore_torch/galore_projector.py	
read	galore_torch/adamw8bit.py	
process	tr \r \n	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python /var/tmp/kv-repro/...	
process	grep -E Eval loss Final eval Saving Total params	
process	tee ../train.out	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/lib/python3.12/site-packages/...	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/lib/python3.12/site-packages/...	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	
process	/var/tmp/kv-repro/033-jiaweizhao-GaLore/env/bin/python -u torchrun_main....	

C Run history

Invocation 1, OK, 1 hours 6 minutes, 22 output lines, started 2026-10-10T19:31:53

metric_hash 8b608fcb9bd79559128b348a5495a2a614cac0e6d31de07bbb656f860034e3b3
run_digest c1f9d4bdbff48b7494c8726c04d65424813a7ded27e2b5e9e09c8ac77c7a287d
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 3ea1024baf8c7e0c27e846061055683df0b20293833dcb6e85c6acd0ddb29262
signed_message_hash 32166d343458d1e4f2df56f934c1376a60e0d9cf53be49c685e385921d5b9347
nonce ab276b13e13cdfb94d4762640f4dfa
signed_at 2026-10-10T20:38:35.345788+00:00
sealed_at 2026-10-10T20:38:35.415785832Z
checkout_bundle 8fff306ee6827640b6a7389baa870efaea748ec278598d61b62383cecd2114f6

Signature (base64)

```
o7Gh1xcyLcj0hn9q9m4aE0ftDRWMjzWUQZKwCoU7mW41mXzZQC0XnB44KUxVyBPVDSda3SirZQl8fA5SZoMTSGgk
0a+qn7uBaeB6xUNNo6ltayi7wD0ks95Cr4rTHzr2d5ingtLy5uYtkkoR9AcksbNJ1ILCixKNLbkWR0ECh5bgBDR
A8uMS36XSrWgmZbCMzycCid15mPkegTKyZUw7SHpxb6m6l0Fscf0PJ+7qnEMFSfpptWvnpqYyhnSpjrHor0l0zal
Gvuz863Cn24NO/U9whMabGegyyw1NR8zUxnT6KfkmGc8kC4PdYmntpUsiYiRUoge8wpgYRPWMof2SAkhvXDJ32Yy
V2GwHEgYbgvpvRUFwKkFmp0gF7SesLTkd0ts6SUW/fWNdsjZcAnzr8fWbwDZtdIXBf8viIuRi1ljXewq8RP0roF
Ecctu0URVuASYyH/AQWrTNeStsY2UGFIi2mMvQBLIiEIZSggnUARvKIULB6mNz1LZ1ty2zphmgzIGX7Rc2zlnOR/
ZRRz8q0wmtc/X5i4+J3HtNgoJ3em2I0h0kQV2irtUlqbbGjckXmLDxT3Xu+oZh3jxQMparCQe/c+jCfILMKTGjS/
u3fx1IzGLInPQKQ92vEOs+JkTgLUwxqMGrpAXPDmsUHehmPXPkoduuvvTyaxA/LDE0Y=
```

Public key

```
-----BEGIN PUBLIC KEY-----
```

```
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAlQL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LoUFz5CUZzRi
6+8FTv70dHuIz5A4Lat4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNPpxYa5X
7qU95EFEEisX/GfoS9ToXtqPBwCcVwSPBX8efec4TC58m96Fsp4mC8KIcEadbIbHN
av0JjdTJ//CMOWZSIbdzE6qvEbNFcqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdhQ+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9IL5xEaJn4bku4eKqeS7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHNcy9bLgPU5annzP9gN2l
6HEMLdkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7LE3mbQRbPsLL/JapDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCkDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdW3gHLCCAWEAAQ==
```

```
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout 19370c226b6375c1037faa782d513ed044da003fa94444f091d385a7865ec6d3
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 8b608fcb9bd79559128b348a5495a2a614cac0e6d31de07bbb656f860034e3b3
digest c1f9d4bdbff48b7494c8726c04d65424813a7ded27e2b5e9e09c8ac77c7a287d
kv_run.sh (unchanged)
pre b0642c4bd9de254310224850094d2e7acd267bedb2a157effbbde45daedd1490
post b0642c4bd9de254310224850094d2e7acd267bedb2a157effbbde45daedd1490
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```